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Chapter 1

Mathematics (Faculty of Science)

1.1 2025 - Problem 1

座標平面上の点 $A(0,0)$, $B(0,1)$, $C(1,1)$, $D(1,0)$ を考える。 $0 < t < 1$ にして, 線分 AB , BC , CD を $t : (1-t)$ に分する点をそれぞれ P_t , Q_t , R_t とし, 線分 P_tQ_t , Q_tR_t を $t : (1-t)$ に分する点をそれぞれ S_t , T_t とする。さらに, 線分 S_tT_t を $t : (1-t)$ に分する点を U_t とする。また, 点 A を U_0 , 点 D を U_1 とする。

(Translation : Consider the points $A(0,0)$, $B(0,1)$, $C(1,1)$, and $D(1,0)$ on the coordinate plane. For a real number $0 < t < 1$, let P_t , Q_t , and R_t be the points dividing the segments AB , BC , and CD internally in the ratio $t : (1-t)$, respectively. Let S_t and T_t be the points dividing P_tQ_t and Q_tR_t internally in the same ratio $t : (1-t)$. Furthermore, let U_t be the point dividing S_tT_t internally in the ratio $t : (1-t)$. Define $U_0 = A$ and $U_1 = D$.)

(1) 点 U_t の座標を求めよ。

(Translation : Find the coordinates of the point U_t .)

(2) t が $0 \leq t \leq 1$ の範を動くときに点 U_t が描く曲線と, 線分 AD でまれた部分の面積を求めよ。

(Translation : Find the area enclosed by the curve traced by U_t as t varies over $0 \leq t \leq 1$ and the line segment AD .)

(3) a を $0 < a < 1$ をたすとする。 t が $0 \leq t \leq a$ の範を動くときに点 U_t が描く曲線の長さを, a の多項式の形で求めよ。

(Translation : Let a be a real number satisfying $0 < a < 1$. As t varies over $0 \leq t \leq a$, find the length of the curve traced by U_t in polynomial form in a .)

1.2 2025 - Problem 2

(1) $x > 0$ のとき, 不等式 $\log x \leq x - 1$ を示せ。

(Translation : For $x > 0$, show that $\ln x \leq x - 1$.)

(2) 次の極限を求めよ。

(Translation : Find the following limit.)

$$\lim_{n \rightarrow \infty} n \int_1^2 \ln \left(\frac{1 + x^{1/n}}{2} \right) dx$$

Remark. In the original problem statement, the logarithm with base e is denoted by $\log x$, but here it is written as $\ln x$ for clarity.

Solution for (1)

Define $h(x) = x - 1 - \ln x$ for $x > 0$.

$$h(1) = 0, \quad h'(x) = 1 - \frac{1}{x}.$$

Then $h'(x) < 0$ on $(0, 1)$ and $h'(x) > 0$ on $(1, \infty)$, hence h decreases on $(0, 1)$ and increases on $(1, \infty)$. Therefore $x = 1$ is the global minimizer of h and

$$h(x) \geq h(1) = 0 \quad (x > 0),$$

i.e.,

$$\boxed{\ln x \leq x - 1 \quad \text{for all } x > 0}$$

with equality iff $x = 1$. Equivalently, one may note $h''(x) = 1/x^2 > 0$, so h is convex and the critical point at $x = 1$ is the global minimum.

Alternative one-line proof for (1)

Since $\ln x$ is a concave function on $(0, \infty)$, its graph lies below any of its tangents. In particular, the tangent at $x = 1$ is

$$y = \ln 1 + (\ln x)'|_{x=1} (x - 1) = x - 1.$$

Hence,

$$\boxed{\ln x \leq x - 1 \quad \text{for all } x > 0,}$$

with equality if and only if $x = 1$.

Solution for (2)

Assume $x \in [1, 2]$. By Arithmetic–Geometric Mean(AM–GM) inequality,

$$\frac{1 + x^{1/n}}{2} \geq \sqrt{x^{1/n}} = x^{1/2n}$$

with equality iff $x = 1$. Using the inequality in (1), we can establish the following inequality,

$$n \ln x^{1/2n} \leq n \ln \left(\frac{1 + x^{1/n}}{2} \right) \leq n \left(\frac{1 + x^{1/n}}{2} - 1 \right)$$

Integrating both sides in $x \in [1, 2]$,

$$n \int_1^2 dx \ln x^{1/2n} \leq n \int_1^2 dx \ln \left(\frac{1 + x^{1/n}}{2} \right) \leq n \int_1^2 dx \left(\frac{1 + x^{1/n}}{2} - 1 \right)$$

1. The leftmost term is computed as

$$n \int_1^2 dx \ln x^{1/2n} = n \left(\frac{1}{2n} \right) \int_1^2 dx \ln x = \frac{1}{2} [x \ln x - x]_1^2 = \ln 2 - \frac{1}{2}$$

2. The rightmost term is computed as

$$\begin{aligned} n \int_1^2 dx \left(\frac{1 + x^{1/n}}{2} - 1 \right) &= \frac{1}{2t} \int_1^2 dx (x^t - 1) \quad \left(\because t = \frac{1}{n} \right) \\ &= \frac{1}{2t} \left[\frac{1}{1+t} x^{t+1} - x \right]_1^2 = \frac{1}{2t} \left[\frac{1}{1+t} (2^{t+1} - 1) - 1 \right] \\ &= \frac{1}{2} \left[\frac{1}{t(1+t)} (2^{t+1} - 2 + 1) - \frac{1}{t} \right] \\ &= \frac{1}{2} \left[2 \times \frac{2^t - 1}{t} \times \frac{1}{1+t} + \frac{1}{t(1+t)} - \frac{1}{t} \right] \\ &= \frac{1}{2} \left[2 \times \frac{2^t - 1}{t} \times \frac{1}{1+t} - \frac{1}{(1+t)t} \right] \end{aligned}$$

As we take the limit $n \rightarrow \infty$, $t \rightarrow 0$, thus

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_1^2 dx \left(\frac{1+x^{1/n}}{2} - 1 \right) &= \lim_{t \rightarrow 0} \frac{1}{2} \left[2 \times \frac{2^t - 1}{t} \times \frac{1}{1+t} - \frac{1}{(1+t)} \right] \\ &= \frac{1}{2} [2 \times \ln 2 \times 10 - 1] = \ln 2 - \frac{1}{2} \end{aligned}$$

3. Now, we take the limit of all terms of our established inequality,

$$\begin{aligned} \lim_{n \rightarrow \infty} n \int_1^2 dx \ln x^{1/2n} &\leq \lim_{n \rightarrow \infty} n \int_1^2 dx \ln \left(\frac{1+x^{1/n}}{2} \right) \leq \lim_{n \rightarrow \infty} n \int_1^2 dx \left(\frac{1+x^{1/n}}{2} - 1 \right) \\ \ln 2 - \frac{1}{2} &\leq \lim_{n \rightarrow \infty} n \int_1^2 dx \ln \left(\frac{1+x^{1/n}}{2} \right) \leq \ln 2 - \frac{1}{2} \end{aligned}$$

Therefore, by the sandwich theorem,

$$\boxed{\lim_{n \rightarrow \infty} n \int_1^2 dx \ln \left(\frac{1+x^{1/n}}{2} \right) = \ln 2 - \frac{1}{2}}$$

1.3 2025 - Problem 3

平行四形 $ABCD$ において, $\angle ABC = \frac{\pi}{6}$, $AB = a$, $BC = b$, $a \leq b$ とする 次の件をたす長方形 $EFGH$ を考え, その面積を S とする .

(Translation : In parallelogram $ABCD$, let $\angle ABC = \pi/6$, $AB = a$, and $BC = b$, where $a \leq b$. Consider a rectangle $EFGH$ satisfying the following conditions, and let its area be S .)

件: 点 A, B, C, D はそれぞれ EF, FG, GH, HE 上にある ただし, はその端の点も含むものとする .

(Translation : Condition: Points A, B, C , and D lie respectively on the sides EF, FG, GH , and HE . Each side includes its endpoints.)

(1) $\angle BCG = \theta$ とするとき, S を a, b, θ を用いて表せ .

(Translation : Let $\angle BCG = \theta$. Express S in terms of a, b , and θ .)

(2) S のとりうる値の最大値を a, b を用いて表せ .

(Translation : Express the maximum possible value of S in terms of a and b .)

1.4 2025 - Problem 4

この問においては, 0 以上の整の 2 になるを平方と呼ぶ a を正の整とし, $f_a(x) = x^2 + x - a$ とおく。

(Translation : In this problem, a square number refers to a number that is the square of a nonnegative integer. Let a be a positive integer, and define $f_a(x) = x^2 + x - a$.)

(1) n を正の整とする $f_a(n)$ が平方ならば, $n \leq a$ であることを示せ。

(Translation : Let n be a positive integer. Show that if $f_a(n)$ is a square number, then $n \leq a$.)

(2) $f_a(n)$ が平方となる正の整 n の個を N_a とおく 次の件 (i), (ii) が同値であることを示せ。

(Translation : Let N_a denote the number of positive integers n such that $f_a(n)$ is a square number. Show that the following two conditions (i) and (ii) are equivalent.)

(i) $N_a = 1$ である。

(Translation : (i) $N_a = 1$.)

(ii) $4a + 1$ は素数である。

(Translation : (ii) $4a + 1$ is a prime number.)

Solution for (1)

Let $f_a(n) = n^2 + n - a = k^2$ for some $k \in N$.

Assume $n > a$. Then

$$n - a = k^2 - n^2 = (k + n)(k - n) > 0 \quad \rightarrow \quad \therefore k > n$$

Now, we can assume $k \geq n + 1$ because $k \in N$. Then

$$\begin{aligned} k^2 &\geq (n + 1)^2 = n^2 + 2n + 1 \\ (n^2 + n - k^2) + n + 1 &\leq 0 \end{aligned} \tag{1}$$

where the inequality (1) leads to a contradiction because

$$n^2 + n - k^2 = a > 0, \quad n + 1 > 0$$

Hence, our original assumption $n > a$ is not valid, as desired.

Solution for (2)

1. Proof from (i) to (ii)

Since $N_a = 1$, let $n \in N$ be a unique positive number such that $f_a(n) = n^2 + n - a = k^2$ for some $k \in N$. Now,

$$\begin{aligned} 4a + 1 &= 4(n^2 + n - k^2) + 1 = (4n^2 + 4n + 1) - 4k^2 \\ &= (2n + 1)^2 - (2k)^2 = (2n + 2k + 1)(2n - 2k + 1) \end{aligned}$$

Let $d_1 = 2n - 2k + 1$ and $d_2 = 2n + 2k + 1$. Then d_1, d_2 are both odd positive integers satisfying $d_2 - d_1 = 4k$.

Assume that $p = 4a + 1$ is not a prime number. Then there exists at least two factorizations $d_1 d_2 = 4a + 1$ such that either $1 < d_1 < d_2$ or $d_1 = 1$.

Case 1. If $d_1 = 1, n = k$.

Case 2. If $1 < d_1 < d_2$, then n, k can be expressed as

$$n = \frac{d_1 + d_2 - 2}{4}, \quad k = \frac{d_2 - d_1}{4}.$$

Now, let us prove that n, k expressed in terms of d_1, d_2 are indeed positive integers.

Recall that $d_2 - d_1 = 4k$ and thus $k \in N$.

Since d_1, d_2 are odd and their product $d_1 d_2 = 4a + 1$ leaves a remainder 1 when divided by 4, both d_1 and d_2 leave the same remainder 1 or 3 upon division by 4. In other words, the possible pairs of d_1, d_2 are $(d_1, d_2) = (4r + 1, 4s + 1)$ or $(d_1, d_2) = (4r + 3, 4s + 3)$ for some integer r, s . In any case, $d_1 + d_2 - 2$ is divisible by 4, and thus $n \in N$. Thus, n, k are both positive integers for any set of d_1, d_2 as defined above.

However,

$$n - k = \frac{1}{4} [(d_1 + d_2 - 2) - (d_2 - d_1)] = \frac{1}{2}(d_1 - 1) > 0$$

Thus, there exist $n \in N$ such that $n > k$. In other words, there exists at least two $n \in N$ such that $f_a(n) = k^2$, which implies $N_a \geq 2$, leading to the contradiction. Hence, our assumption that $p = 4a + 1$ is not a prime number is false, as desired.

2. Proof from (ii) to (i)

Assume that $p = 4a + 1$ is a prime number and let $k \in \mathbb{N}$ such that $f_a(n) = k^2$. As shown above,

$$p = 4a + 1 = (2n + 2k + 1)(2n - 2k + 1).$$

Let $d_1 = 2n - 2k + 1$ and $d_2 = 2n + 2k + 1$. Then d_1, d_2 are both odd positive integers satisfying $d_2 - d_1 = 4k$. Since p is prime, the only possible factorization is

$$(d_1, d_2) = (1, p).$$

Hence,

$$n = \frac{d_1 + d_2 - 2}{4} = a, \quad k = \frac{d_2 - d_1}{4} = a.$$

Therefore, n is uniquely determined by the factorization $p = d_1 d_2$, implying $N_a = 1$.

Caution : Wrong proof from (i) to (ii)

Since $N_a = 1$, let $n \in N$ be a unique positive number such that $f_a(n) = n^2 + n - a = k^2$ for some $k \in N$. Now,

$$\begin{aligned} 4a + 1 &= 4(n^2 + n - k^2) + 1 \\ &= (4n^2 + 4n + 1) - 4k^2 \\ &= (2n + 1)^2 - (2k)^2 = (2n + 2k + 1)(2n - 2k + 1) \end{aligned}$$

Assume $(4a + 1)$ is not a prime number. Then $2n - 2k + 1 > 1$, implying $n > k$. By our proof for (1), $n \leq a$ and thus $n \leq k$ ($\because n - a = (k + n)(k - n)$), which leads to contradiction. Thus our original assumption that $(4a + 1)$ is not a prime number is false, as desired.

Why is this wrong? The conclusion that “if $(4a + 1)$ is not a prime number, then $2n - 2k + 1 > 1$ ” is not logically sufficient or necessary.

The fact that $(4a + 1)$ is composite only guarantees the existence of *some* nontrivial factorization, but it does not exclude the possibility that, in this particular factorization $(2n + 2k + 1)(2n - 2k + 1)$, the smaller factor could still be 1.

Therefore, the subsequent inference $n > k$ lacks a valid logical basis.

1.5 2025 - Problem 5

n を 2 以上の整数とする。1 から n までの字が書かれた札が各 1 枚ずつ合計 n 枚あり、一列におかれている。1 以上 $(n-1)$ 以下の整数 i にして、次の操作 (T_i) を考える。

(Translation : Let n be an integer greater than or equal to 2. There are n cards, each labeled with one of the numbers from 1 to n , arranged in a horizontal row. For each integer i satisfying $1 \leq i \leq n-1$, consider the following operation (T_i) .)

(T_i) 左から i 番目の札の字が、左から $(i+1)$ 番目の札の字よりも大きければ、これら 2 枚の札の位置を入れかえる。そうでなければ、札の位置をかえない。

(Translation : (T_i) If the number on the i -th card from the left is greater than the number on the $(i+1)$ -th card, swap these two cards. Otherwise, leave them as they are.)

最初の態において札の字は左から $A_1, A_2, \dots, A_n$ であったとする。この態から $(n-1)$ 回の操作 $(T_1), (T_2), \dots, (T_{n-1})$ を順に行った後、けて $(n-1)$ 回の操作 $(T_{n-1}), \dots, (T_2), (T_1)$ を順に行ったところ、札の字は左から $1, 2, \dots, n$ と小さい順に並んだ。以下の問いに答えよ。

(Translation : Initially, the numbers on the cards from left to right are $A_1, A_2, \dots, A_n$. Starting from this state, perform the sequence of $(n-1)$ operations $(T_1), (T_2), \dots, (T_{n-1})$ in order, and then another $(n-1)$ operations $(T_{n-1}), \dots, (T_2), (T_1)$ in reverse order. After these operations, the cards are arranged in increasing order from left to right: $1, 2, \dots, n$. Answer the following questions.)

(1) A_1 と A_2 のうち少なくとも一方は 2 以下であることを示せ。

(Translation : Show that at least one of A_1 or A_2 is less than or equal to 2.)

(2) 最初の態にしてありうる札の字のび方 $A_1, A_2, \dots, A_n$ のを c_n とする。 n が 4 以上の整数であるとき、 c_n を c_{n-1} と c_{n-2} を用いて表せ。

(Translation : Let c_n be the number of possible initial arrangements $A_1, A_2, \dots, A_n$ of the cards. For integers $n \geq 4$, express c_n in terms of c_{n-1} and c_{n-2} .)

1.6 2025 - Problem 6

複素平面上の点 $\frac{1}{2}$ を中心とする半 $\frac{1}{2}$ の円の周から原点を除いた曲線を C とする。

(Translation : Let C be the curve obtained by removing the origin from the circumference of the circle centered at the point $\frac{1}{2}$ with radius $\frac{1}{2}$ in the complex plane.)

(1) 曲線 C 上の複素 z にし、 $\frac{1}{z}$ の部は 1 であることを示せ。

(Translation : Show that for any complex number z on the curve C , the real part of $\frac{1}{z}$ is 1.)

(2) α, β を曲線 C 上の相異なる複素とすると、 $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ がとりうる範囲を複素平面上に示せよ。

(Translation : Let α and β be distinct complex numbers on the curve C . Illustrate on the complex plane the region of possible values of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.)

(3) γ を (2) で求めた範囲にさない複素とすると、 $\frac{1}{\gamma}$ の部がとりうる値の最大値と最小値を求めよ。

(Translation : Let γ be a complex number not belonging to the region found in (2). Determine the maximum and minimum possible values of the real part of $\frac{1}{\gamma}$.)

Chapter 2

Mathematics (Faculty of Humanities and Social Sciences)

2.1 2025 - Problem 1

a を正のとする 座標平面において, 放物線 $C: y = x^2$ 上の点 $P(a, a^2)$ における C の接線と直交し, P を通る直線を ℓ とおく ℓ と C の交点のうち, P と異なる点を Q とおく .

(Translation : Let a be a positive real number. In the coordinate plane, let ℓ be the line passing through the point $P(a, a^2)$ on the parabola $C: y = x^2$ and perpendicular to the tangent to C at P . Let Q be the point of intersection of ℓ and C other than P .)

(1) Q の x 座標を求めよ .

(Translation : Find the x -coordinate of Q .)

Q における C の接線と直交し, Q を通る直線を m とおく m と C の交点のうち, Q と異なる点を R とおく .

(Translation : Let m be the line passing through Q and perpendicular to the tangent to C at Q . Let R be the point of intersection of m and C other than Q .)

(2) a がすべての正のを動くとき, R の x 座標の最小値を求めよ .

(Translation : As a varies over all positive real numbers, find the minimum value of the x -coordinate of R .)

2.2 2025 - Problem 2

平面上で $AB = AC = 1$ である二等三角形 ABC を考える。正の r にし、 A, B, C それぞれを中心とする半 r の円3つを合わせた領域を D_r とする。ただし、この問いでは、三角形と円は周とその部からなるものとする。° AB, AC, BC がすべて D_r に含まれるような最小の r を s 、三角形 ABC が D_r に含まれるような最小の r を t と表す。°

(Translation : In the plane, consider an isosceles triangle ABC with $AB = AC = 1$. For a positive real number r , let D_r denote the union of three circles each of radius r centered at A, B , and C . In this problem, both the triangle and circles include their interiors and boundaries. Let s be the smallest r such that all sides AB, AC , and BC are contained in D_r , and let t be the smallest r such that the entire triangle ABC is contained in D_r .)

(1) $\angle BAC = \frac{\pi}{3}$ のとき、 s と t を求めよ。°

(Translation : Find s and t when $\angle BAC = \frac{\pi}{3}$.)

(2) $\angle BAC = \frac{2\pi}{3}$ のとき、 s と t を求めよ。°

(Translation : Find s and t when $\angle BAC = \frac{2\pi}{3}$.)

(3) $0 < \theta < \pi$ をたす θ にして、 $\angle BAC = \theta$ のとき、 s と t を θ を用いて表せ。°

(Translation : For $0 < \theta < \pi$, express s and t in terms of θ when $\angle BAC = \theta$.)

2.3 2025 - Problem 3

白玉2個がにんでいる 投げたとき表と裏の確率がそれぞれ $\frac{1}{2}$ のコインを用いて, 次の手順(*)をくり返し, 白玉または玉を一行にべていく。

(Translation : Two white balls are placed side by side. Using a fair coin with heads and tails each having probability $\frac{1}{2}$, repeat the following procedure (*), placing white or black balls in a horizontal row.)

手順(*) コインを投げ, 表がでたら白玉, 裏がでたら玉を, それまでにべられている一番右にある玉の右隣におく。そして, 新しくおいた玉の色がその1つ左の玉の色と異なり, かつ2つ左の玉の色と一致するときには, 新しくおいた玉の1つ左の玉を新しくおいた玉と同じ色の玉にとりかえる。

(Translation : (*) Toss the coin. If heads appears, place a white ball; if tails, place a black ball to the right of the rightmost existing ball. If the newly placed ball differs in color from the ball immediately to its left but matches the color of the ball two positions to its left, then replace the ball immediately to the left with a ball of the same color as the newly placed one.)

例えば, 手順(*)を2回行いコインが裏, 表の順にでた場合には, 白玉が4つぶ。

(Translation : For example, if the coin shows tails then heads in two repetitions of procedure (*), four white balls will be lined up.)

正の整 n にして, 手順(*)を n 回行った時点での $(n+2)$ 個の玉のび方を考える。

(Translation : For a positive integer n , consider the arrangement of $(n+2)$ balls after performing procedure (*) n times.)

(1) $n = 3$ のとき, 右から2番目の玉が白玉である確率を求めよ。

(Translation : Find the probability that the second ball from the right is white when $n = 3$.)

(2) n を正の整とする 右から2番目の玉が白玉である確率を求めよ。

(Translation : Let n be a positive integer. Find the probability that the second ball from the right is white.)

(3) n を正の整数とする 右から1番目と2番目の玉がともに白玉である確率を求めよ。

(Translation : Let n be a positive integer. Find the probability that both the first and second balls from the right are white.)

2.4 2025 - Problem 4

a をとする 座標平面において, 次の連立不等式の表す領域の面積を $S(a)$ とする .

(Translation : Let a be a real number. In the coordinate plane, let $S(a)$ denote the area of the region represented by the following system of inequalities:)

$$\begin{cases} y \leq -\frac{1}{2}x^2 + 2, \\ y \geq |x^2 + a|, \\ -1 \leq x \leq 1 \end{cases}$$

a が $-2 \leq a < 2$ の範を動くとき, $S(a)$ の最大値を求めよ .

(Translation : As a varies over $-2 \leq a < 2$, find the maximum value of $S(a)$.)